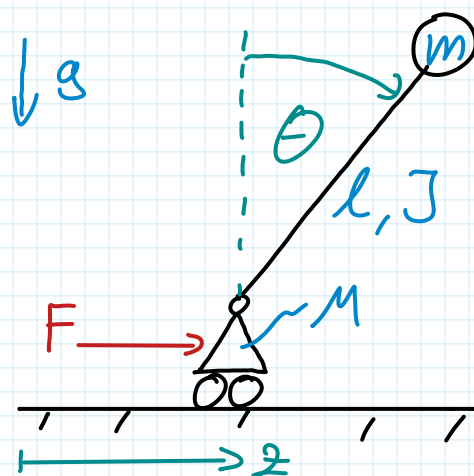


• Bigger picture:

1) We want to control a system...



Inverted Pendulum

So we find the DE's:

$$\begin{aligned}\dot{x} &= f(x, u) \\ y &= g(x, u)\end{aligned}$$

$$\begin{aligned}\cdot x &= (z \quad \dot{z} \quad \theta \quad \dot{\theta})^T \\ \cdot u &= F \\ \cdot y &= \theta\end{aligned}$$

2) Linearize around an equilibrium point...

• Find Jacobi-matrices and plug in.

$$x_e = (0 \ 0 \ 0 \ 0)^T, \quad u_e = 0$$

$$\begin{pmatrix} \dot{x} \\ \ddot{x} \\ \dot{\theta} \\ \ddot{\theta} \end{pmatrix} = \underbrace{\begin{pmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \dots & a_{nn} \end{pmatrix}}_A \begin{pmatrix} x \\ \dot{x} \\ \theta \\ \dot{\theta} \end{pmatrix} + \underbrace{\begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}}_B u$$

$$y = \underbrace{(C_1 \dots C_n)}_C \begin{vmatrix} x \\ \dot{x} \\ \theta \\ \dot{\theta} \end{vmatrix} + \underbrace{d}_{\tilde{D}} \cdot u$$

3) With A, B, C, D and the initial condition of the states x_0 , we can find $y(t)$ using...

$$y(t) = \underbrace{C e^{At} x_0}_{\text{Initial condition response}} + \underbrace{\int_0^t e^{A(t-\tau)} B u(\tau) d\tau + D u(t)}_{\text{Forced response}}$$

This is where we left off. We simplified the problem using diagonalization of the A matrix and can now tackle control... Or can we???

• 2 problems that we still have:

1) The integral in the solution

$$y(t) = Ce^{At}x_0 + \int_0^t e^{A(t-\tau)} Bu(\tau) d\tau + Du(t)$$

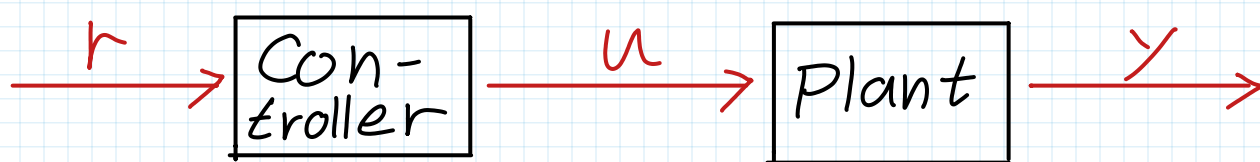
$\Rightarrow$ This is actually a convolution!

• Def: $(f * g) = \int_0^t f(\tau)g(t-\tau)d\tau$

$\Rightarrow$ Convolutions are computationally intense

2) Connecting signals

• There will be multiple blocks!



$\Rightarrow$ Input u to the plant is the output of the controller, so to find y , we must evaluate

$$u(t) = C_2 e^{A_1 t} x_0 + \int_0^t e^{A_1(t-z)} B_1 r(z) dz + D_1 r(t),$$

and only then we can find

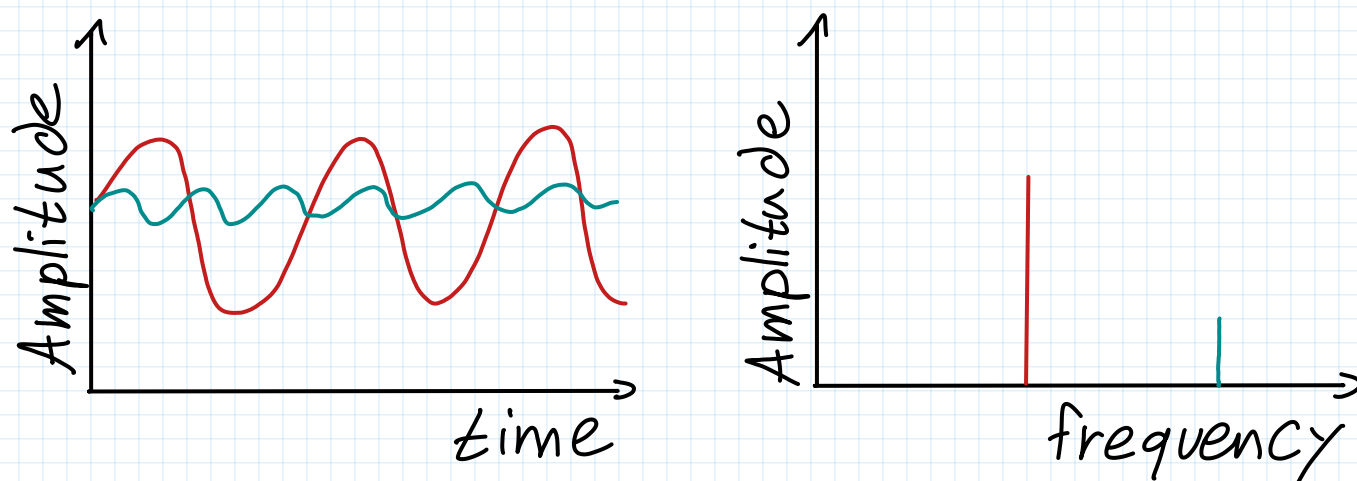
$$y(t) = C_2 e^{A_2 t} x_0 + \int_0^t e^{A_2(t-z)} B_2 u(z) dz + D_2 u(t).$$

$\Rightarrow$ This is not very efficient, and there is a very clever way to tackle these problems.

Things will get a little abstract now, it's okay if you don't immediately have an intuition.

Frequency Domain

- General idea: Instead of looking how a system behaves over time, we look at how it behaves over different frequencies.
- This means we represent signals differently:



How do we get here?

⇒ We use the Laplace-transform:

$$\mathcal{L}\{f\}(s) = \int_0^{\infty} f(t) e^{-st} dt$$

• you don't need to solve this, you can use tables:

$x(t)$	$X(s)$
impulse: $\delta(t)$	1
step: $h(t)$	$\frac{1}{s}$
$h(t) \cdot t^n$	$\frac{n!}{s^{n+1}}$
$h(t) \cdot e^{at}$	$\frac{1}{s-a}$
$h(t) \cdot t^n \cdot e^{at}$	$\frac{n!}{(s-a)^{n+1}}$
$h(t) \cdot \sin(\omega \cdot t)$	$\frac{\omega}{s^2 + \omega^2}$
$h(t) \cdot \cos(\omega \cdot t)$	$\frac{s}{s^2 + \omega^2}$
$h(t) \cdot \sinh(\omega \cdot t)$	$\frac{\omega}{s^2 - \omega^2}$
$h(t) \cdot \cosh(\omega \cdot t)$	$\frac{s}{s^2 - \omega^2}$
$k \cdot u(t-a)$	$k \cdot \frac{e^{-as}}{s}, a > 0$
$\delta(t-a)$	$e^{-as}, a > 0$

• The Laplace transform also has the following identity:

$$\mathcal{L}\{f^{(n)}(t)\} = s^n F(s)$$

Which we can use to transform our systems.

$$\dot{x}(t) = Ax(t) + Bu(t) \quad (I)$$

$$y(t) = Cx(t) + Du(t) \quad (II)$$

$\Rightarrow$ Apply $\mathcal{L}\{\}$ to (I):

$$\mathcal{L}\{\dot{x}(t)\} = \mathcal{L}\{Ax(t) + Bu(t)\}$$

$$sX(s) = AX(s) + BU(s)$$

$$(sI - A)X(s) = BU(s)$$

$$X(s) = (sI - A)^{-1}BU(s) \quad (III)$$

=> Now transform (II)

$$\mathcal{L}\{y(t)\} = \mathcal{L}\{Cx(t) + Du(t)\}$$

$$Y(s) = CX(s) + DU(s) \quad (\text{IV})$$

=> plug (III) into (IV)

$$Y(s) = C(sI - A)^{-1}BU(s) + DU(s)$$

$$Y(s) = (C(sI - A)^{-1}B + D)U(s)$$

• From this, we can define the next, very important concept: The Transfer Function.

$$G(s) = \frac{Y(s)}{U(s)} = C(sI - A)^{-1}B + D$$

=> It describes the ratio of **output** to **input** of a system in frequency domain!

Let's try an example...

Exam HS2022

Problem: You are given a linear time-invariant system of the form:

$$\dot{x}(t) = A \cdot x(t) + B \cdot u(t),$$

$$y(t) = C \cdot x(t) + D \cdot u(t).$$

} state-space
⇒ time domain

Q16 (1 Points)

Given:

$$A = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}, B = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, C = [0 \quad 1], D = -1.$$

Which of the following transfer functions $G(s)$ is equivalent to the given state space system. Mark the correct answer.

☐ $G(s) = -\frac{s^2 - 3s + 2}{s^2 - 4s + 1}$

☐ $G(s) = -\frac{(s+2) \cdot (s-1)}{s^2 - 4s + 1}$

☐ $G(s) = \frac{s^2 - 3s + 2}{s^2 - 4s + 1}$

☐ $G(s) = -\frac{s^2 + 3s + 2}{s^2 - 4s + 1}$

} transfer-functions
⇒ frequency domain

• Apply the formula $G(s) = C(sI - A)^{-1}B + D$:

$$G(s) = (0 \quad 1) \begin{pmatrix} s-1 & 1 \\ 2 & s-3 \end{pmatrix}^{-1} \begin{pmatrix} 1 \\ -1 \end{pmatrix} - 1$$

• The inverse can be time-intensive...
⇒ Use this identity:

$$A^{-1} = \frac{\text{adj}(A)}{\det(A)}$$

With $\text{adj}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix},$

and $\det\begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - cb.$

$$\Rightarrow \begin{pmatrix} s-1 & 1 \\ 2 & s-3 \end{pmatrix}^{-1} = \frac{1}{(s-1)(s-3)-2} \begin{pmatrix} s-3 & -1 \\ -2 & s-1 \end{pmatrix}$$

$$G(s) = \frac{1}{(s-1)(s-3)-2} (0 \ 1) \begin{pmatrix} s-3 & -1 \\ -2 & s-1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} - 1$$

$$= \frac{1}{(s-1)(s-3)-2} (-2 \ s-1) \begin{pmatrix} 1 \\ -1 \end{pmatrix} - 1$$

$$= \frac{-s-1}{(s-1)(s-3)-2} - 1 = \frac{-s-1-(s-1)(s-3)+2}{(s-1)(s-3)-2}$$

$$= \frac{-s^2+3s-2}{s^2-4s+1} = - \frac{s^2-3s+2}{s^2-4s+1}$$

- ☒ $G(s) = -\frac{s^2-3s+2}{s^2-4s+1}$
- ☐ $G(s) = -\frac{(s+2) \cdot (s-1)}{s^2-4s+1}$
- ☐ $G(s) = \frac{s^2-3s+2}{s^2-4s+1}$
- ☐ $G(s) = -\frac{s^2+3s+2}{s^2-4s+1}$

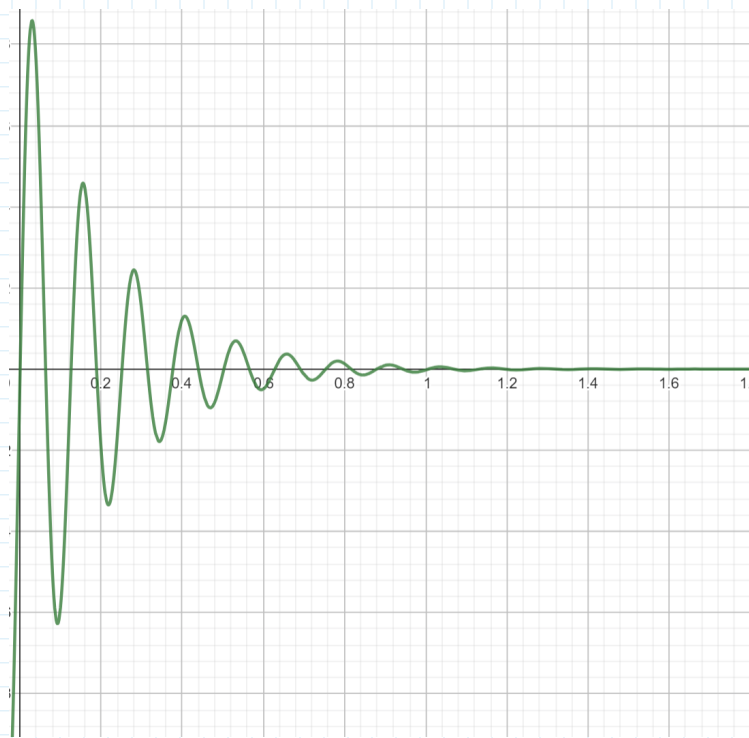
1P!

😊

So we now know how to get to the frequency domain. But what really is a transfer function?
And what really is "s"?

- Remember the purpose of the Laplace-Transform...

=> The Laplace-Transform describes a function as a combination of sinusoids and exponentials only.



$$f(t) = e^{-5t} \sin(50t)$$

We can describe many signals in this form!

=> Visualization

<https://johnflux.com/2019/02/12/laplace-transform-visualized/>

- So let's look at a general signal:

$$f(t) = 1e^{\sigma t} \cos(\omega t)$$

- Rewrite using $\cos(x) = \frac{1}{2}(e^{jx} + e^{-jx})$

Euler's Formula

$$\Rightarrow f(t) = \frac{1}{2} e^{\sigma t} (e^{j\omega t} + e^{-j\omega t})$$

$$f(t) = \frac{1}{2} (e^{(\sigma + j\omega)t} + e^{(\sigma - j\omega)t})$$

- The signal is fully characterized by the exponent $\sigma \pm j\omega$, and the magnitude 1
- Due to linearity, we can now write general signals as

$$f_{\text{gen}}(t) = \sum_i \lambda_i e^{\sigma_i \pm j\omega_i t}$$

Let's take a closer look at $\sigma + j\omega$.

- $\sigma + j\omega$ has the structure of a complex number $a + jb$. The complex frequency is named:

$$s = \sigma + j\omega.$$

$$\Rightarrow \text{Re}(s) = \sigma : \text{exponential part}$$

$$\text{Im}(s) = \omega : \text{sinusoidal part}$$

- If we now want to know, what a system $G(s)$ does with some complex frequency, we just plug in...

Ex:

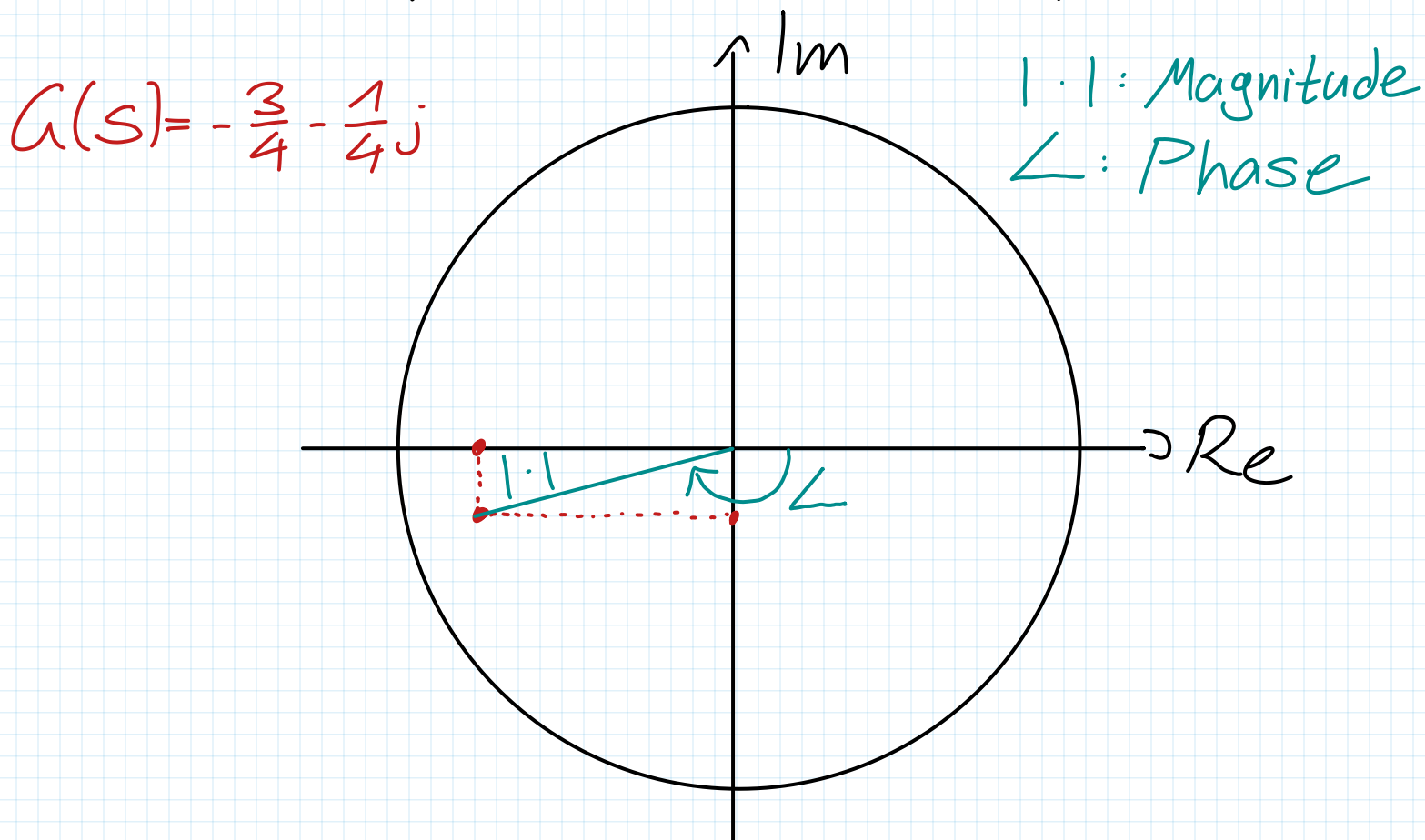
Given a pure sinusoid with $\omega = 1 \frac{\text{rad}}{\text{s}}$, what does $G(s)$ do with it?

$$G(s) = -\frac{s^2 - 3s + 2}{s^2 - 4s + 1} \quad \left| \begin{array}{l} \text{Plug in } s = \sigma + j\omega \\ s = 1j \frac{\text{rad}}{\text{s}} \end{array} \right.$$

$$= -\frac{-1 - 3j + 2}{-1 - 4j + 1} = \frac{1 - 3j}{4j} = -\frac{3}{4} - \frac{1}{4}j$$

So we just get another complex number?

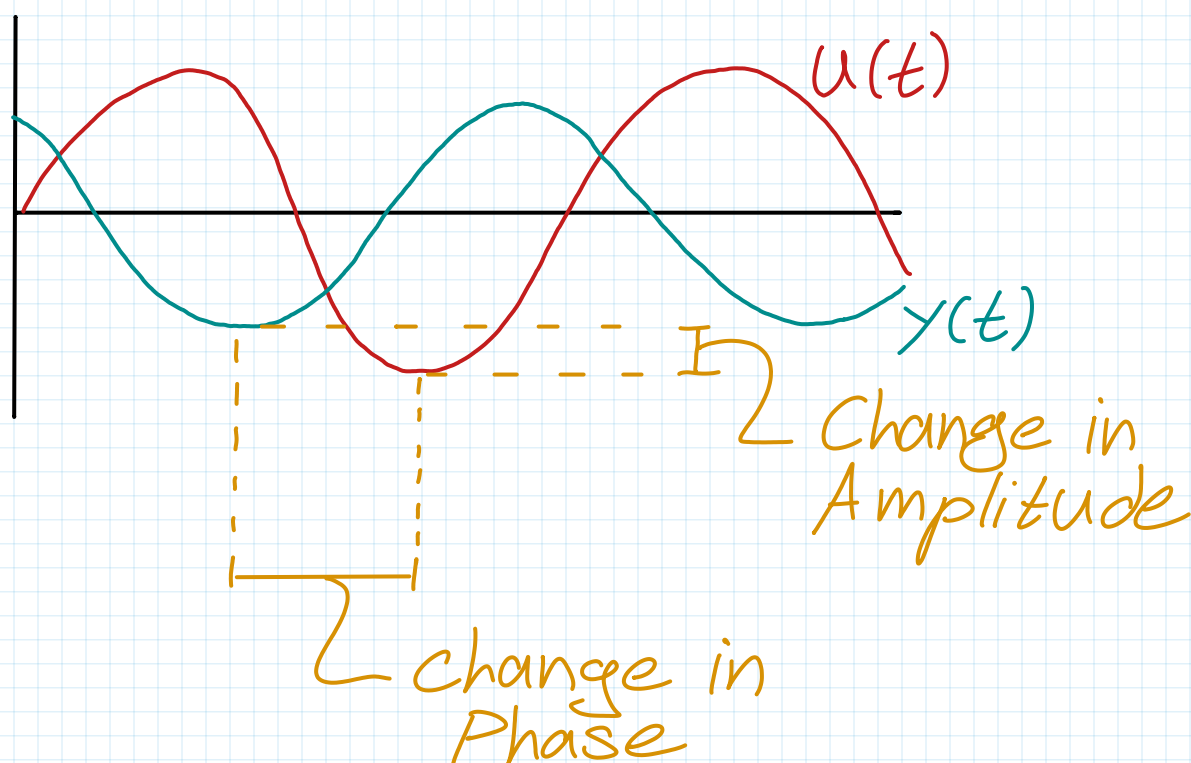
$\Rightarrow$ Yes! From which we can extract the amplitude and the phase.



• Magnitude: $|- \frac{3}{4} - \frac{1}{4}j| = \sqrt{(-\frac{3}{4})^2 + (-\frac{1}{4})^2} \approx 0,79$

• Phase: $\angle -\frac{3}{4} - \frac{1}{4}j = \text{atan2}\left(\frac{-1/4}{-3/4}\right) \approx -2,8 \text{ rad}$

A seemingly random number and some angle? What does this mean?



- Remember, our $G(s)$ is a fraction!

$$G(s) = \frac{Y(s)}{U(s)} = -\frac{s^2 - 3s + 2}{s^2 - 4s + 1}$$

$\Rightarrow$ It shows us the ratio of **output** to **input**.

- $|G(s)|$: ratio of amplitudes
- $\angle G(s)$: phase change

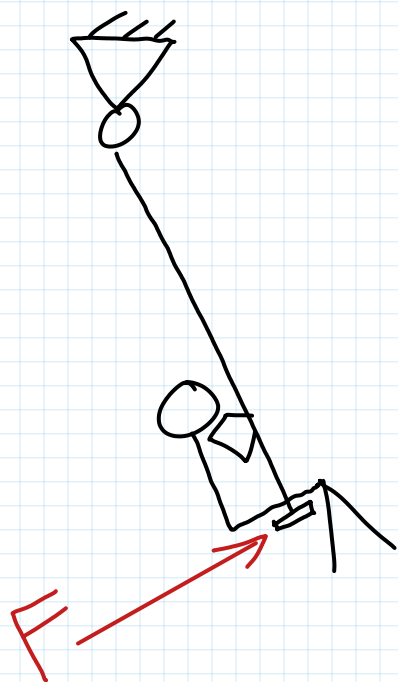
Let's recap...

- Many systems are excited with sinusoidal frequencies. With a transfer function, we can find out how the system warps the amplitude and phase of a system.

=> And the frequency?

- The output of a LTI-system, always has the same frequency as the input!

Intuition:



- $F_1(t) = \cos(1000t)$

- $F_2(t) = \cos(0,001t)$

=> You must hit the right frequency to strongly amplify the amplitude.

Now we start to see the use of the frequency domain.

$\Rightarrow$ But theres more!

1) The integral in the solution

$$y(t) = Ce^{At}x_0 + \int_0^t e^{A(t-z)}Bu(z)dz + Du(t)$$

$\Rightarrow$ This is actually a convolution!

• Def: $(f * g) = \int_0^t f(z)g(t-z)dz$

$\Rightarrow$ Convolutions are computationally intense

2) Connecting signals

• There will be multiple blocks!



• These are the problems we still had in time domain.

The Laplace transform has one more identity that solves these issues!

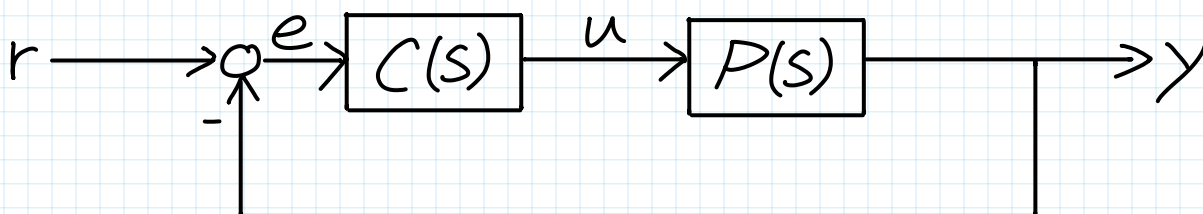
• Convolution Theorem:

$$\mathcal{L}\{f(t) * g(t)\} = F(s) \cdot G(s)$$

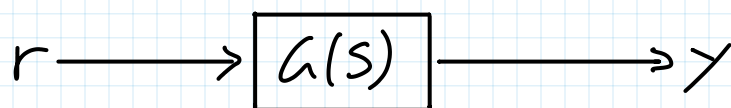
$\Rightarrow$ So the hard to solve integral becomes a simple multiplication.

• Also, if we have a connection of blocks, we can evaluate the total system by simply multiplying the transfer functions.

Ex: $C(s) = \frac{1}{s+2}$, $P(s) = \frac{s+2}{(s-1)(s+1)}$



we already saw, that we can rewrite:



where $G(s) = \frac{C(s)P(s)}{1 + C(s)P(s)}$

so here $G(s) = \frac{\cancel{\frac{1}{s+2}} \cdot \cancel{\frac{s+2}{(s-1)(s+1)}}}{1 + \cancel{\frac{1}{s+2}} \cdot \cancel{\frac{s+2}{(s-1)(s+1)}}}$

$$= \frac{\frac{1}{(s-1)(s+1)}}{\frac{(s-1)(s+1) + 1}{(s-1)(s+1)}} = \cancel{\frac{1}{s^2}}$$

$\Rightarrow$ Entering frequency domain is really useful. But what if have a tf given and need to go back to time-domain?

- We can find one possible realization with this formula:

Given:

$$G(s) = \frac{b_{n-1}s^{n-1} + b_{n-2}s^{n-2} + \dots + b_0}{s^n + a_{n-1}s^{n-1} + \dots + a_0} + d$$

Then:

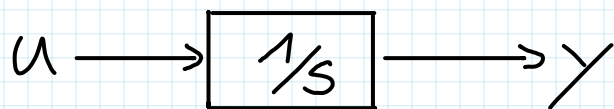
$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & \dots & 0 \\ 0 & 0 & 1 & 0 & \dots & 0 \\ \vdots & \vdots & & \ddots & & \vdots \\ -a_0 & -a_1 & \dots & & & -a_{n-1} \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

$$C = [b_0 \ b_1 \ \dots \ b_{n-1}], \quad D = [d]$$

$\Rightarrow$ This realization is called the controllable canonical form

That's all the theory for today. Let's look at some quiz questions...

Q1) What does the following tf do?



Answer: $y = \int u dt$

$$\begin{cases} \dot{x} = u \\ y = x \end{cases} \quad \} \quad y(t) = x(t) = \int u(t) dt$$

$\Rightarrow$ write as $\dot{x} = ax + bu$, $y = cx + du$

$\Rightarrow a=0, b=1, c=1, d=0$

Find TF as: $G(s) = C(sI - A)^{-1}B$

$$= 1 \cdot (s - 0)^{-1} \cdot 1 = \underline{\underline{1/s}}$$

Q2) The transfer function of any system can be written as a ratio of two polynomials in s .

☐ TRUE ☒ FALSE

$\Rightarrow$ Must be SISO-LTI

Q3) Exam HS22

Q17 (1 Points)

Given:

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ -2 & 0 & -4 & -1 & -6 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}, C = \begin{bmatrix} \frac{1}{2} & -2 & 0 & -\frac{1}{3} & 0 \end{bmatrix}, D = 0.$$

Which of the following transfer functions $G(s)$ is equivalent to the given state space system. Mark the correct answer.

- ☐ $G(s) = \frac{\frac{1}{2}s^3 - 2s + \frac{1}{3}}{s^5 + 2s^4 + 4s^3 + 1s^2 + 6}$
- ☒ $G(s) = \frac{-\frac{1}{2}s^3 - 2s + \frac{1}{3}}{s^5 + 6s^4 + s^3 + 4s^2 + 2}$
- ☐ $G(s) = \frac{-\frac{1}{3}s^3 - 2s + \frac{1}{2}}{s(s^4 + 6s^3 + s^2 + 4s + 2)}$
- ☐ $G(s) = \frac{s(\frac{1}{2}s^3 - 2s + \frac{1}{3})}{s^5 + 2s^4 + 4s^3 + 1s^2 + 6}$

Given:

$$G(s) = \frac{b_{n-1}s^{n-1} + b_{n-2}s^{n-2} + \dots + b_0}{s^n + a_{n-1}s^{n-1} + \dots + a_0} + d$$

Then:

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & \dots & 0 \\ 0 & 0 & 1 & 0 & \dots & 0 \\ \vdots & \vdots & & \ddots & & \vdots \\ -a_0 & -a_1 & \dots & & -a_{n-1} \end{bmatrix}, B = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

$$C = [b_0 \ b_1 \ \dots \ b_{n-1}], \quad D = [d]$$